

Full paired invariant-mixture analysis

Dense branch grid; 20,000 null and 20,000 alternative replicates per model $\times$ branch $\times$ alignment-length cell

Legacy counterfactual

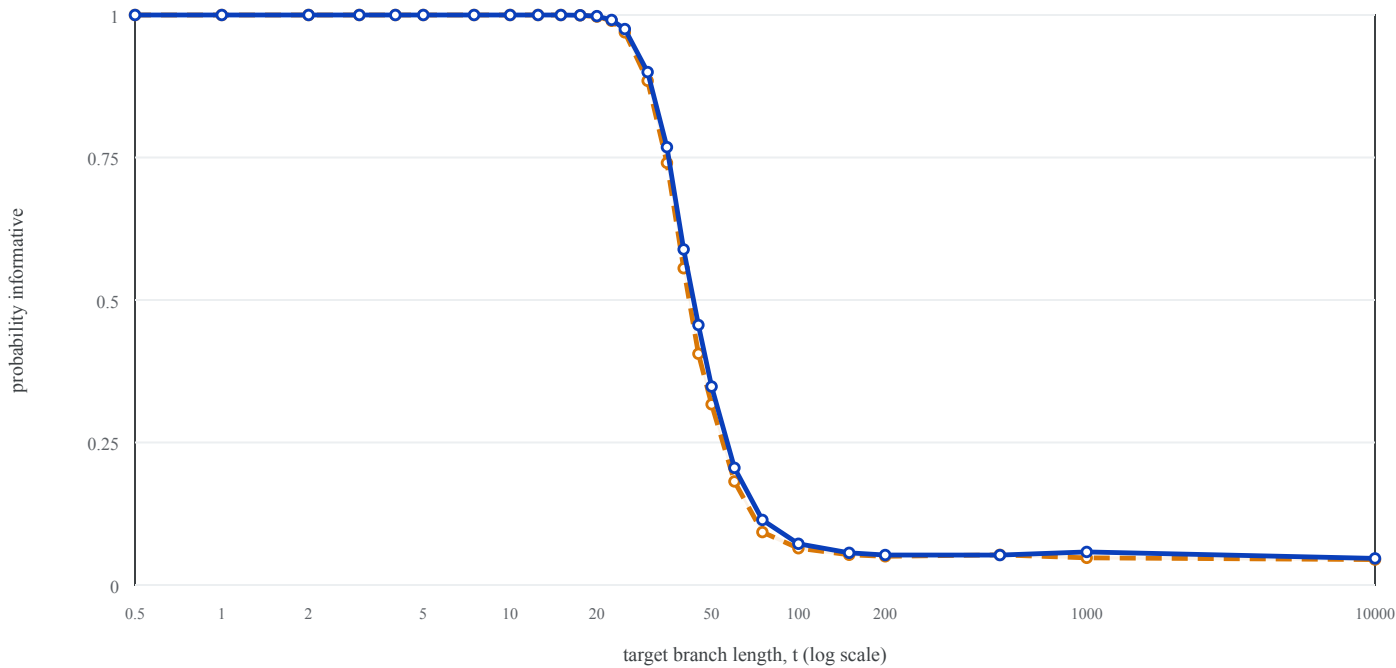
Refined invariant null

+I+G4

+I+R4

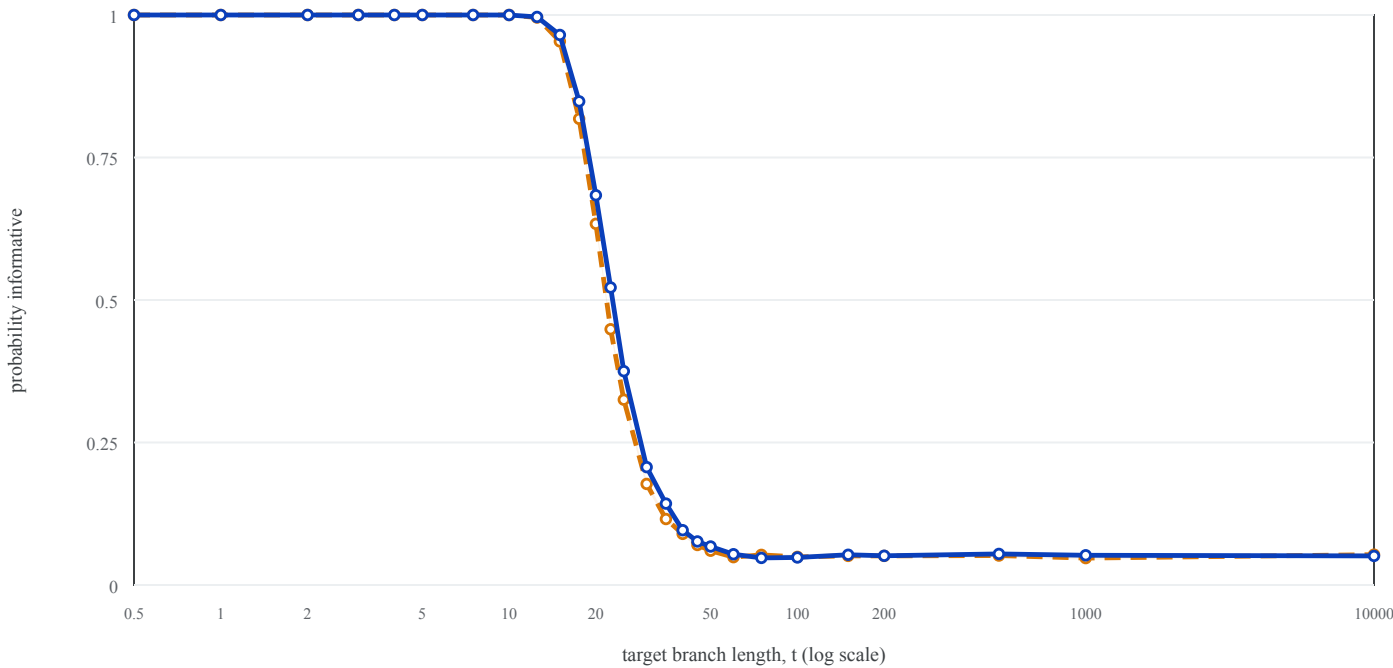
A +I+G4 calibrated power

n = 1,000; bands are Wilson 95% intervals



B +I+R4 calibrated power

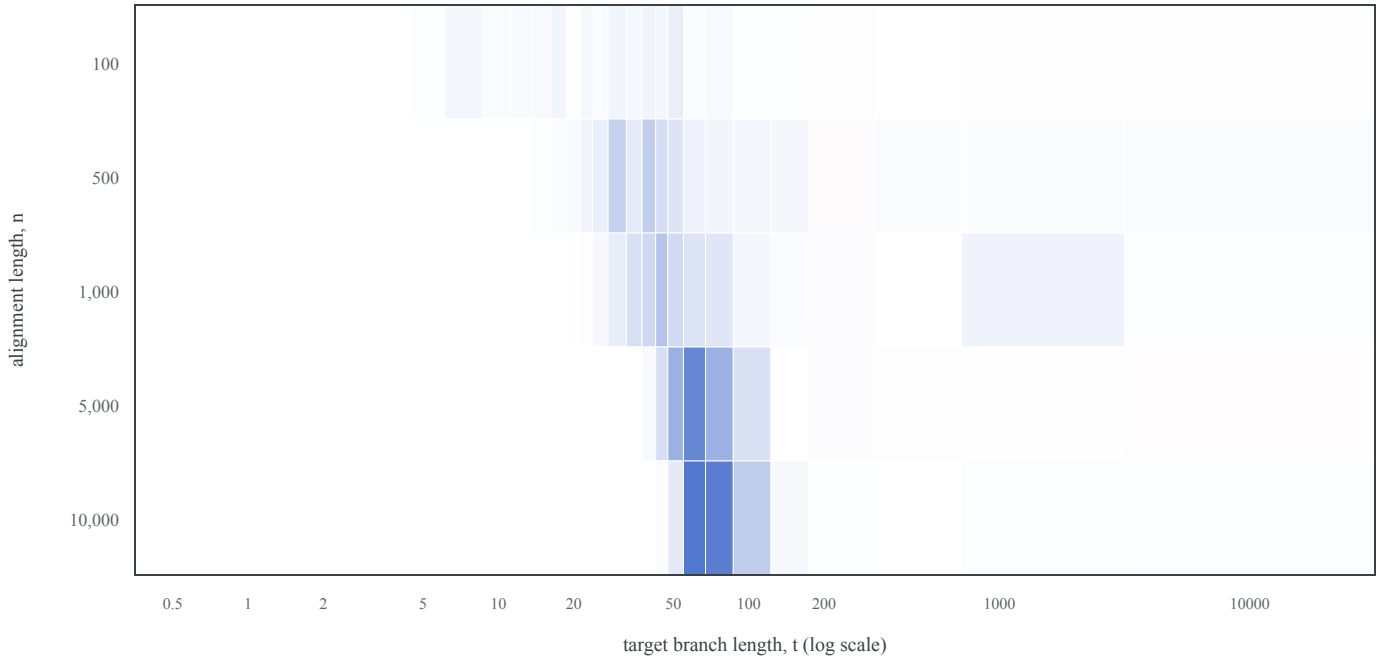
n = 1,000; bands are Wilson 95% intervals



C +I+G4 paired power difference

refined minus legacy after independent null calibration

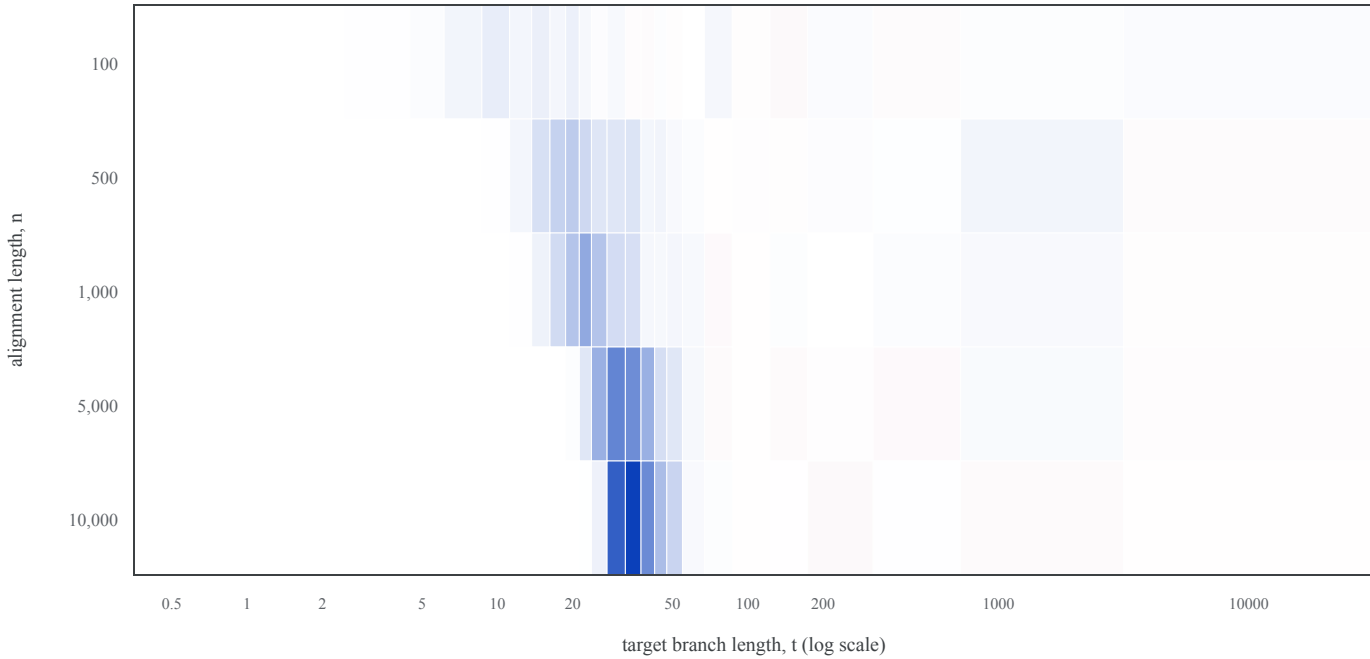
-0.164 0 +0.164



D +I+R4 paired power difference

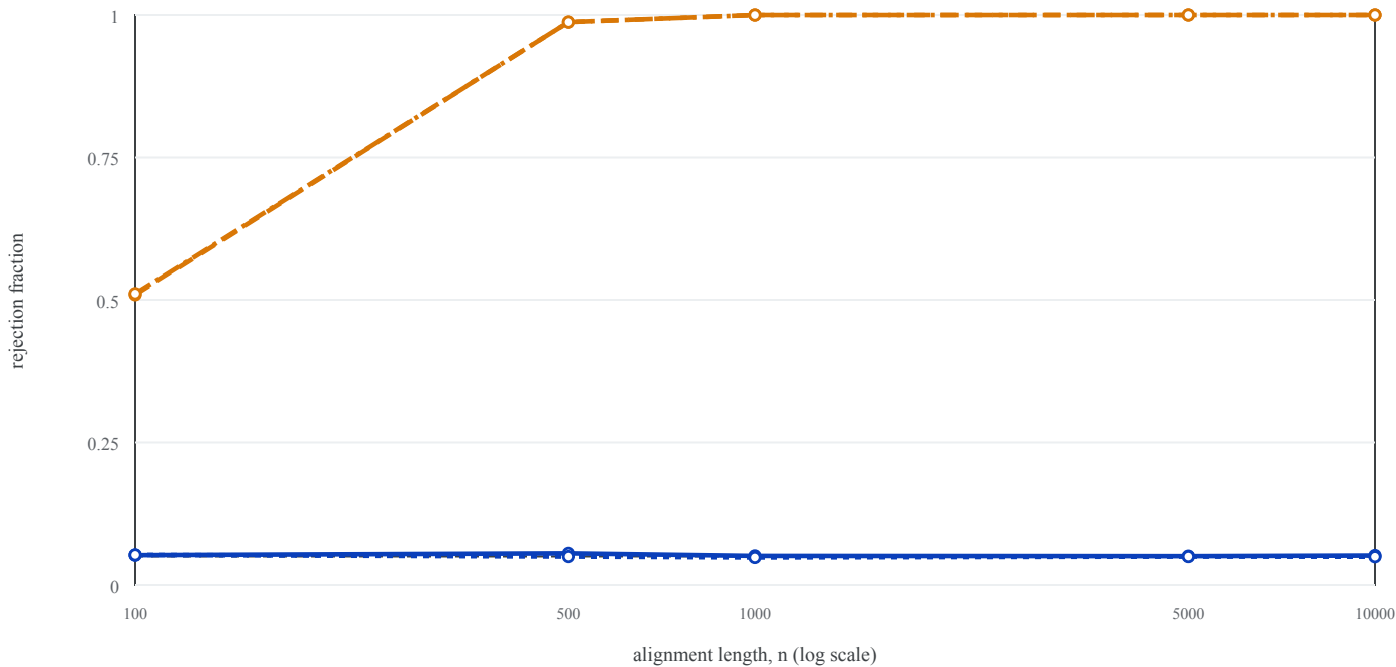
refined minus legacy after independent null calibration

-0.164 0 +0.164



E Extreme-branch nominal false positives

t = 10,000 under the correct invariant-mixture null



F Extreme-branch empirical thresholds

95th percentile trained on an independent half-null sample

